

Quadratic Simultaneous Equations

AQA Higher · GCSE Grade 8–9

Name: _____ Class: _____ Date: _____

What you need to be able to do

Solve a pair of equations where one is a straight line (linear) and one is a curve (quadratic), using substitution. Most pairs have two answer points — find both x values and the y that goes with each.

Common mistakes to avoid

- Finding only one answer — most pairs have two solution points.
- Forgetting to set the quadratic equal to 0 before factorising.
- Sign slips when expanding brackets like $(x - 1)^2$ or moving terms across.
- Solving for x but forgetting to find the matching y.

Method steps

1. Step 1: Make y (or x) the subject of the straight-line equation.
2. Step 2: Substitute that into the quadratic equation.
3. Step 3: Move everything to one side so it equals 0.
4. Step 4: Solve the quadratic — factorise or use the formula.
5. Step 5: Put each value back to find its pair. Write answers as (x, y).

Worked example

Solve: $y = x^2 - 2x + 3$ and $y = x + 3$

Substitute: $x^2 - 2x + 3 = x + 3$

Rearrange: $x^2 - 3x = 0$

Factorise: $x(x - 3) = 0$ $x = 0$ or $x = 3$

Find y: $x = 0$ $y = 3$; $x = 3$ $y = 6$

Answers: (0, 3) and (3, 6)

Question 1

[3 marks] ●○○○○

Solve the simultaneous equations.

$y = x^2$ and $y = 2x + 3$

x = _____

y = _____

Solve the simultaneous equations.

$y = x^2 + 1$ and $y = x + 3$

x =

y =

Question 3

[3 marks] ●●○○○

Solve the simultaneous equations.

$$y = x^2 - 2x \text{ and } y = x + 4$$

x = _____

y = _____

Question 4

[3 marks] ●●○○○

Solve the simultaneous equations.

$$y = x^2 - 5x + 6 \text{ and } y = x - 2$$

x = _____

y = _____

Question 5

[4 marks] ●●●○○

Solve the simultaneous equations.

$$x^2 + y^2 = 25 \text{ and } y = x - 1$$

x = _____

y = _____

Question 6

[4 marks] ●●●○○

Solve the simultaneous equations.

$$x^2 + y^2 = 10 \text{ and } x + y = 4$$

x = _____

y = _____

Question 7**[5 marks]** ●●●●○

Solve the simultaneous equations. Give your answers in surd form.

$$x^2 + y^2 = 9 \text{ and } y = x + 1$$

x = _____**y** = _____

Question 8**[5 marks]** ●●●●●

A rectangle has an area of 12 cm^2 and a perimeter of 14 cm .

By forming and solving a pair of simultaneous equations, find the length and width of the rectangle.

(Let the length be x and the width be y .)

x = _____

y = _____

Self-reflection

How confident do you feel with this topic now? (tick one)

- ☐ Not yet ☐ Getting there ☐ Confident ☐ Really confident

Which question did you find the hardest, and why?

One method step I want to remember next time:

One thing I will practise before the next lesson:

Teacher key

1. Q1: (3, 9) and (-1, 1)
2. Q2: (2, 5) and (-1, 2)
3. Q3: (4, 8) and (-1, 3)
4. Q4: (4, 2) and (2, 0)
5. Q5: (4, 3) and (-3, -4)
6. Q6: (1, 3) and (3, 1)
7. Q7: $x = (-1 + \sqrt{17})/2$, $y = (1 + \sqrt{17})/2$ and $x = (-1 - \sqrt{17})/2$, $y = (1 - \sqrt{17})/2$
8. Q8: length 4 cm, width 3 cm (from $xy = 12$ and $x + y = 7$)